

NLP-Driven Patient Risk Stratification in Long-Term Care Using Unstructured Nursing Notes

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Abstract

This study develops an NLP-driven patient risk stratification system for a long-term care facility using one year of electronic health record data comprising 215,460 nursing notes, 36,513 diagnosis records, and 30,744 medication records across 2,153 unique patients. The original objective was to predict dehydration risk, but exploratory data analysis revealed severe class imbalance — only five unique patients (0.20%) were ever labeled as dehydrated — making supervised classification infeasible. The project pivoted to an NLP-based risk scoring approach grounded in published clinical indicators of dehydration in elderly patients. Nursing note text was analyzed using weighted keyword extraction across three severity tiers, and patients on ten or more concurrent medication classes received an additional polypharmacy risk weight. Each of 6,845 patient assessments was classified as STABLE, MONITOR, ELEVATED, or HIGH RISK, with risk trajectories tracked across consecutive assessments. A logistic regression model achieved an F1 score of 0.903 and AUC-ROC of 0.985, substantially outperforming a polypharmacy-only baseline (F1 = 0.593). Nursing note keyword content was the dominant predictor of HIGH RISK classification. This study demonstrates that meaningful risk stratification is achievable from unstructured clinical text even without a reliable labeled outcome variable.

Keywords: natural language processing, patient risk stratification, long-term care, electronic health records, nursing notes, keyword extraction, polypharmacy, logistic regression, clinical dashboard

Introduction

Long-term care facilities serve some of the most medically complex patients in the healthcare system — elderly residents with multiple chronic conditions, high medication burdens, and significant functional dependencies. These facilities generate large volumes of clinical data every day, including nursing progress notes, diagnosis records, and medication logs. Despite the richness of this data, it is rarely analyzed systematically to identify patients at risk of deterioration. Clinical staff must rely on manual observation and judgment, which is time-consuming, inconsistent, and difficult to scale across a large patient population (Van Bulck et al., 2025; Li et al., 2023).

In recent years, natural language processing (NLP) has emerged as a promising approach for extracting clinical insights from unstructured text in electronic health records (EHRs). Spasic and Nenadic (2020) conducted a systematic review of clinical text data in machine learning, documenting a wide range of NLP methods for processing nursing notes, discharge summaries, and physician documentation. Van Bulck et al. (2025) demonstrated that NLP can unlock clinically actionable patterns from EHR free text in real-world nursing settings. At the same time, machine learning models have been applied to structured clinical data — including diagnosis codes and medication records — to predict patient outcomes and flag at-risk populations (Qu et al., 2025; Acharya et al., 2024).

Nursing notes in particular have received growing research attention as a data source for patient risk prediction. Mitha et al. (2023) reviewed 43 studies applying NLP to nursing-generated notes, finding that they contain clinical signals not captured in structured data. Reunamo et al. (2025) showed that supervised ML applied to nursing notes could identify patient needs in pediatric oncology settings. Weng et al. (2025) found that sentiment polarity in nursing notes predicted perioperative complications, demonstrating that the language nurses use carries prognostic value. In intensive care settings, Pilowsky et al. (2024) found that combining nursing notes with structured EHR data consistently improved predictive performance over using either source alone.

Despite this progress, several gaps remain in the literature. Most NLP studies in clinical settings focus on acute care or specific disease prediction tasks with well-labeled outcome data (Spasic & Nenadic, 2020; Pilowsky et al., 2024). Long-term care facilities, which have distinct documentation practices and patient populations, have received comparatively little attention. Additionally, most existing approaches rely on sufficient labeled training data — a requirement that is often unmet in real-world healthcare datasets where target outcomes may be extremely rare (Luu et al., 2024). Kansal et al. (2021) highlighted that even seemingly straightforward choices such as how to group ICD-10 diagnosis codes can significantly affect model performance, yet these decisions are rarely documented or justified in clinical ML studies. Finally, the question of how to present risk insights to clinicians in an interpretable and actionable format remains underexplored (Mohindra et al., 2024; Bucalon et al., 2022).

This study addresses these gaps by developing an NLP-driven patient risk scoring system for a long-term care facility, using one year of EHR data comprising 215,460 nursing notes, 36,513

diagnosis records, and 30,744 medication records across 2,153 patients. Rather than requiring a well-labeled outcome variable, the approach uses weighted keyword extraction from nursing note text, combined with a polypharmacy flag derived from medication data, to assign risk scores to each patient assessment. Risk trajectories are tracked over time to identify patients whose condition is worsening. Results are presented through an interactive Python-based dashboard designed for clinical usability.

This study is guided by three hypotheses. First, patients whose nursing notes contain higher frequencies of negative clinical keywords — such as “refused,” “confused,” “urine,” and “fall” — will have systematically higher risk scores than patients with predominantly positive or neutral notes. Second, patients on ten or more concurrent medication classes (polypharmacy) will show elevated risk scores compared to patients on fewer medications. Third, risk scores will exhibit meaningful temporal trends within individual patients, with patients classified as worsening showing increasing scores across consecutive assessments.

The contribution of this study is twofold. Methodologically, it demonstrates that meaningful patient risk stratification is achievable from nursing note text even in the absence of a balanced labeled outcome variable — a common real-world constraint. Practically, the resulting risk scoring system and interactive dashboard provide a prototype tool for long-term care facilities to systematically identify and monitor high-risk patients, supporting more proactive and data-driven clinical care (Mesinovic et al., 2025; Steiger & Kroll, 2023).

Literature Review

Clinical Risk Prediction Using Machine Learning and NLP

A substantial body of research has applied machine learning to clinical prediction tasks using structured EHR data. Qu et al. (2025) compared multiple ML models for predicting dehydration risk in patients with acute diarrhea, finding that model choice significantly affected discriminative performance for ordinal outcomes. Their work provides a direct methodological benchmark for clinical prediction tasks of the type attempted in this project. However, a recurring challenge across clinical ML studies is class imbalance — when the outcome of interest is rare, standard models tend to classify nearly all cases as negative and achieve misleadingly high accuracy. Luu et al. (2024) addressed this directly, providing a practical guide to handling imbalanced clinical datasets using techniques such as SMOTE oversampling, class weighting, and precision-recall metrics. These considerations are directly relevant to this project, where fewer than 0.2% of patient assessments carry a positive dehydration label.

Alongside structured data approaches, NLP has emerged as a complementary method for extracting risk signals from unstructured clinical text. Spasic and Nenadic (2020) conducted a systematic review of clinical text data in machine learning, cataloguing preprocessing methods, feature extraction strategies, and model architectures across a wide range of clinical NLP tasks. Their findings underscore both the potential and the complexity of working with free-text clinical documentation. Van Bulck et al. (2025) extended this work to nursing-specific contexts, showing that NLP applied to EHR text can support direct clinical decision-making rather than

just research applications. Acharya et al. (2024) examined clinical risk prediction using large language models (LLMs), noting that LLMs offer particular advantages when labeled training data is scarce — which is precisely the constraint this project faces.

NLP Applied to Nursing Notes

Nursing notes present unique challenges and opportunities for NLP. Mitha et al. (2023) reviewed 43 NLP studies applied specifically to nursing-generated documentation, finding that nurse-written notes differ linguistically from physician notes — they tend to be shorter, more fragmented, and more varied in structure. The authors identified keyword extraction and rule-based methods as the most common approaches, alongside more sophisticated methods such as named entity recognition and sentiment analysis. These findings directly informed the preprocessing strategy used in this project, particularly the decision to join fragmented note rows and use weighted keyword scoring.

Reunamo et al. (2025) demonstrated the clinical value of nursing note ML in a pediatric oncology setting, where supervised models trained on note text identified psychosocial support needs not captured in structured data. While the patient population differs from the long-term care context of this project, the core finding — that nursing notes contain clinically meaningful signal beyond what structured fields record — applies broadly. Weng et al. (2025) added a sentiment dimension to this picture, showing that the emotional tone of nursing documentation correlates with clinical outcomes including postoperative complications and length of stay. In intensive care settings, Pilowsky et al. (2024) found that NLP applied to clinical notes, when combined with structured data sources, consistently outperformed either source alone — a finding that motivates this project's multi-source approach integrating notes, diagnoses, and medications.

Diagnosis Code Handling and Feature Engineering

One underappreciated methodological challenge in clinical ML is the handling of ICD-10 diagnosis codes. Raw ICD-10 datasets typically contain hundreds or thousands of unique codes, most of which appear too rarely to be useful as individual features. Kansal et al. (2021) studied this problem systematically, comparing multiple code grouping strategies — including raw codes, truncated codes, and established groupers such as CCS and the Elixhauser comorbidity index — across two large hospital datasets. Their finding that grouping method significantly affects model performance has direct implications for this project, which works with 875 unique diagnosis codes across 2,153 patients. Steiger and Kroll (2023) proposed an alternative approach using patient embeddings derived from diagnosis code profiles, representing each patient as a vector in a learned diagnostic space. Their Pat2Vec framework offers a potential avenue for future feature engineering in this project, particularly if the analysis expands beyond keyword-based risk scoring.

Clinical Dashboards for Decision Support

The translation of ML outputs into clinically usable tools is an active area of research. Mohindra et al. (2024) described the development and implementation of an EHR-integrated dashboard for oncology shared decision-making, reporting that clinician-centered design, multiple data sources,

and interpretable visualizations were critical to adoption. Bucalon et al. (2022) conducted a scoping review of 18 clinical dashboard studies, identifying key design principles including real-time data integration, role-appropriate information presentation, and actionable alert design. Both studies inform the dashboard component of this project, which aims to present patient risk scores and trends in a format that supports clinical prioritization. Mesinovic et al. (2025) addressed the specific challenge of explainability in LLM-based clinical tools, arguing that transparent, step-by-step reasoning is essential for clinical trust — a consideration relevant to any future LLM integration in this project.

Transition to the Current Study

Taken together, the literature establishes that nursing notes contain meaningful clinical signal, that NLP methods can extract this signal effectively, and that the resulting insights can be presented to clinicians through well-designed dashboards. However, most existing work focuses on acute care settings with sufficient labeled outcome data. This study addresses the gap by applying NLP-based risk scoring to a long-term care dataset where the labeled outcome is too sparse for supervised learning. By combining keyword extraction from nursing notes with polypharmacy flags from medication data and temporal trend detection across assessment dates, this project develops a risk stratification approach suited to the real-world constraints of long-term care clinical data.

Methods

1. Overview

To develop a patient risk stratification system for a long-term care facility, a multi-component NLP-based approach was employed. The methodology combined weighted keyword extraction from unstructured nursing note text, a polypharmacy flag derived from structured medication records, and temporal trend detection across assessment dates. A logistic regression model was subsequently trained to identify the features most predictive of high-risk patient classifications, and the results were presented through an interactive Streamlit dashboard. Because the dehydration target exhibited severe class imbalance, an NLP-based risk scoring system was developed as a more clinically meaningful alternative to supervised binary classification.

2. Data

Data were obtained from a long-term care facility database through a healthcare data analytics company. The dataset covered the 2019 calendar year and comprised three CSV files: a diagnosis file containing ICD-10 diagnosis codes, a medication file containing medication category records, and a nursing notes file containing free-text progress notes written by nursing staff. Each file included a binary dehydration target variable and an assessment date linking records across files. All data were anonymized at source with resident IDs replacing patient names. All analyses were conducted in Python using Google Colab.

3. Preprocessing and Cleaning

Date columns across all three files were parsed to datetime format. In the diagnosis file, missing diagnosis codes were identified and confirmed to be randomly distributed across months, then

filled with the label 'UNKNOWN' to preserve row count. In the nursing notes file, rows containing an invalid target value were removed, and rows with missing note text were filled with empty strings. Nursing notes were stored in fragments due to a database storage constraint, so all note fragments were joined per patient per assessment date into a single full-text record. ICD-10 diagnosis codes were mapped to readable category names using a custom lookup dictionary. Medication data were merged by patient ID only, rather than by patient and assessment date jointly, since nursing notes and medication records follow different assessment schedules for the same patient.

4. Feature Engineering

NLP risk scoring: A weighted keyword extraction approach was applied to each patient assessment's joined note text. A clinical vocabulary of 29 risk-associated keywords was defined, organized into three severity tiers based on established clinical indicators of dehydration and deterioration in elderly patients (Ogbolu et al., 2025; Li et al., 2023): high-weight keywords directly associated with dehydration such as 'dehydrat', 'urine', 'poor intake', 'skin turgor', and 'dry mouth'; medium-weight keywords indicating general clinical decline such as 'refused', 'confused', 'fall', 'fever', and 'pain'; and low-weight keywords indicating mild concern such as 'dry', 'fluid', 'nausea', and 'wound'. Stabilizing keywords such as 'stable', 'alert', and 'good intake' were assigned negative weights to reduce scores for assessments documenting positive clinical status. Keyword matching used substring search to capture all morphological variants. The risk score was computed as the sum of weights, floored at zero, and assessments were classified into four categories: STABLE, MONITOR, ELEVATED, and HIGH RISK. This tiered scoring structure follows the precedent of validated clinical early warning scores such as the Modified Early Warning Score (MEWS), which similarly sums weighted clinical indicators and stratifies patients into risk tiers (Santiago González et al., 2023).

Polypharmacy flag: Patients on ten or more concurrent medication classes were flagged as polypharmacy cases, contributing additional weight to the final risk score. This threshold reflects established clinical definitions of polypharmacy in elderly populations (Li et al., 2023).

Trend detection: Risk trajectories were calculated for patients with two or more assessments by comparing consecutive risk scores. An increase of more than two points was classified as WORSENING; a decrease of more than two points as IMPROVING; all others as STABLE.

Model features: Eight features were constructed for the logistic regression model: count of risk keywords, polypharmacy flag, number of medication classes, number of unique diagnoses, binary flags for dehydration-related keywords, refusal keywords, and fall-related keywords, and the patient's risk score from the prior assessment.

5. Model Selection

A logistic regression model was selected as the primary approach because the target is binary, the feature set is small and interpretable, and the primary goal was to understand which factors drive high-risk classifications rather than maximize predictive accuracy. The model was implemented using scikit-learn with the lbfgs solver, L2 regularization ($C=1.0$), a maximum of 1,000 iterations, and `class_weight='balanced'` to address class imbalance. Logistic regression

coefficients directly quantify the contribution of each feature to the probability of a HIGH RISK classification, providing an interpretable answer to the research question: 'What are the major drivers of high risk?' A random forest classifier was trained on the same feature set as a non-linear alternative to compare feature importance estimates across modelling approaches.

6. Training and Validation

The dataset was split into training and test sets using stratified random sampling to preserve the class distribution across splits. Class imbalance was addressed by setting `class_weight='balanced'` in both models, which adjusts the loss function to penalize minority class errors more heavily (Luu et al., 2024). Features were standardized using z-score normalization prior to logistic regression fitting. A baseline model predicting HIGH RISK based on the polypharmacy flag alone was used as the performance benchmark. Random seeds were set to 42 for reproducibility.

7. Evaluation Metrics

The primary evaluation metric was F1 score, chosen because the clinical cost of missing a high-risk patient outweighs the cost of a false positive alert, and because accuracy is misleading under class imbalance. Area under the ROC curve (AUC-ROC) was reported as a secondary metric to assess overall discriminative ability. Both metrics were calculated on the held-out test set.

8. Dashboard

An interactive patient risk dashboard was developed using Streamlit, a Python-based web application framework. The dashboard displays population-level visualizations including risk category distribution, patient trend distribution, and average risk score over time. It also provides a per-patient lookup interface showing each patient's current risk score and category, clinical keywords detected in their most recent assessment, medication and diagnosis profile, a risk score timeline across all assessments, and a preview of their nursing notes. The dashboard was designed to support clinical prioritization by enabling staff to filter patients by risk category or trend direction (Bucalon et al., 2022; Mohindra et al., 2024).

9. Tools and Environment

All analyses were conducted in Python using Google Colab. Key packages included pandas for data manipulation, scikit-learn for modelling, matplotlib and seaborn for visualization, and Streamlit and Plotly for dashboard development. All random operations used `random_state=42` for reproducibility. Code and outputs are available in the shared Google Drive project folder.

Results

1. Overview of Results

This study aimed to determine whether nursing note text, medication burden, and diagnosis records from a long-term care EHR system could be used to stratify patients by clinical risk level, and to identify which features most strongly predict high-risk patient classifications. An NLP-based keyword scoring system was applied to 6,845 patient assessments to generate risk scores, and a logistic regression model was trained to predict HIGH RISK assessments using eight engineered features. The most important finding was that nursing note content —

specifically the number and type of clinical risk keywords — was the dominant predictor of high-risk classification, accounting for the majority of the model's predictive power. The logistic regression model achieved an F1 score of 0.903 and AUC-ROC of 0.985 on the held-out test set.

2. Dataset Characteristics

The final analytic dataset comprised 36,513 diagnosis records, 30,744 medication records, and 215,460 nursing note rows for 2,153 unique patients. After joining note fragments per patient per assessment date, 6,845 unique patient-assessment combinations were available for analysis. The dataset was split into 5,476 training assessments (80%) and 1,369 test assessments (20%), with the HIGH RISK class representing 36.2% of both splits, confirming the stratified sampling preserved class distribution. Of the 2,153 patients, 534 had only one assessment on record. Missing diagnosis codes (1,719 records, 4.7%) were filled with 'UNKNOWN' following confirmation of random distribution. Table 1 (Appendix) summarizes the analytic dataset.

3. Exploratory Data Analysis

The dehydration target variable was severely imbalanced: only five unique patients were ever labeled as dehydrated across all three files, representing 0.20% of unique assessments in the diagnosis file, 0.17% in the medication file, and 0.15% in the notes file (Figure 1, Appendix). This level of imbalance precluded supervised dehydration prediction and motivated the pivot to NLP-based risk scoring.

The most common diagnoses in the population were Muscle Disorder (M62, 2,220 records), Cognitive/Memory Symptoms (R41, 1,556), Abnormal Gait/Mobility (R26, 1,374), and Hypertension (I10, 1,362) (Figure 2, Appendix). The most frequently prescribed medication categories were Chemicals (1,979 records), Antidepressants (1,204), and Dermatologicals (1,165) (Figure 3, Appendix). The average patient had 17.3 unique diagnosis codes and 16.5 unique medication classes on record.

Keyword frequency analysis revealed substantial differences between dehydrated and non-dehydrated assessments. The keyword 'urine' appeared in 90.0% of dehydrated assessments compared to 18.2% of non-dehydrated assessments (71.8 percentage-point difference). Additional keywords showing elevated frequency in dehydrated assessments included 'dry' (50.0% vs. 29.2%), 'water' (40.0% vs. 23.0%), 'output' (20.0% vs. 3.6%), and 'hydrat' (20.0% vs. 9.3%) (Figure 4, Appendix).

4. Feature Engineering and Selection

Eight features were constructed for the predictive model. NLP-derived features included: num_risk_keywords (count of negative clinical keywords), has_dehydration_kw (binary flag for 'urine'/'dehydrat'), has_refusal_kw (binary flag for 'refused'), and has_fall_kw (binary flag for 'fall'/'fell'). Structured features included: polypharmacy_flag (binary: 10+ medication classes), num_medications (total unique medication classes), num_diagnoses (total unique diagnoses), and prior_risk_score (risk score from the previous assessment, set to 0 for first assessments). No dimensionality reduction was applied given the small feature set. All features were standardized to zero mean and unit variance prior to logistic regression fitting.

5. Predictive Model Performance

The logistic regression model achieved an F1 score of 0.903 and AUC-ROC of 0.985 on the test set, with precision of 0.89 and recall of 0.91 for the HIGH RISK class and overall accuracy of 0.93. The random forest classifier achieved an F1 score of 0.887, AUC-ROC of 0.981, precision of 0.84, recall of 0.93, and accuracy of 0.91. The polypharmacy-only baseline achieved an F1 score of 0.593 and AUC-ROC of 0.622 (Table 2, Appendix).

6. Comparative Analysis of Models

Both the logistic regression and random forest models substantially outperformed the polypharmacy-only baseline across all metrics. The logistic regression model outperformed the random forest on F1 score (0.903 vs. 0.887) and AUC-ROC (0.985 vs. 0.981), while the random forest achieved higher recall (0.93 vs. 0.91) with lower precision (0.84 vs. 0.89). The baseline model showed high recall (0.94) but very low precision (0.43), indicating that polypharmacy alone over-predicts HIGH RISK. No cross-validation was performed; performance estimates are based on a single stratified 80/20 train-test split (Figure 5, Appendix).

7. Model Interpretation

Logistic regression feature coefficients showed that `num_risk_keywords` had the largest positive coefficient (4.92), followed by `has_dehydration_kw` (1.32) and `polypharmacy_flag` (1.31). The features `has_refusal_kw` (0.63), `has_fall_kw` (0.62), and `prior_risk_score` (0.20) showed smaller positive coefficients. The features `num_diagnoses` (-0.08) and `num_medications` (-0.12) showed small negative coefficients (Figure 6, Appendix). Random forest feature importance confirmed the dominance of NLP features: `num_risk_keywords` accounted for 59.4% of total importance, `has_dehydration_kw` for 16.8%, `has_fall_kw` for 5.1%, `has_refusal_kw` for 5.0%, `prior_risk_score` for 4.9%, `num_medications` for 4.8%, `polypharmacy_flag` for 3.0%, and `num_diagnoses` for 1.1%.

8. Validation and Robustness

The model was evaluated on a held-out test set of 1,369 assessments that were not seen during training. The stratified split preserved the class distribution (36.2% HIGH RISK) in both training and test sets, ensuring comparability. The consistency of results between the logistic regression and random forest models — two distinct algorithms with different assumptions — provides additional evidence of robustness. Patients with only one assessment on record (534 patients) were included in the analysis with `prior_risk_score` set to 0. Their mean risk score (5.70) and HIGH RISK rate (34.1%) were nearly identical to patients with multiple assessments (5.73 and 36.4% respectively), confirming that their inclusion did not introduce systematic bias.

9. Error Analysis

In the logistic regression model test set results, 874 assessments were true non-HIGH RISK and 495 were true HIGH RISK. Of the 495 true HIGH RISK assessments, 45 were misclassified as non-HIGH RISK (false negatives, 9.1%). Of the 874 true non-HIGH RISK assessments, 52 were misclassified as HIGH RISK (false positives, 5.9%). The random forest produced 35 false negatives (7.1%) and 80 false positives (9.2%), showing a different error profile — fewer missed

HIGH RISK patients but more false alarms. In a clinical context, false negatives (missed high-risk patients) are the more costly error type, which favors the logistic regression model's performance profile.

10. Subgroup Analysis

Medication profiles differed between dehydrated and non-dehydrated patient subgroups. Among dehydrated patients, the most common medication categories were Chemicals, Diuretics, Hematopoietic Agents, Minerals & Electrolytes, and Beta Blockers. Among non-dehydrated patients, the most common categories were Chemicals, Antidepressants, Dermatologicals, Laxatives, and Anticonvulsants (Figure 7, Appendix). The presence of Diuretics and Minerals & Electrolytes in the dehydrated subgroup is consistent with active fluid management in these patients.

11. Summary of Findings

The NLP-based keyword scoring system assigned risk scores ranging from 0 to 23 across 6,845 patient assessments, with 23.7% classified as HIGH RISK, 24.0% ELEVATED, 34.2% MONITOR, and 18.1% STABLE. Trend detection classified 319 patients as WORSENING, 269 as IMPROVING, and 859 as STABLE. The logistic regression model trained on eight features achieved an F1 score of 0.903 and AUC-ROC of 0.985, outperforming both the random forest (F1 = 0.887) and the polypharmacy-only baseline (F1 = 0.593). Nursing note content — specifically the number and type of clinical risk keywords — was the strongest predictor of HIGH RISK classification across both models, accounting for 59.4% of random forest feature importance and producing the largest logistic regression coefficient (4.92). The polypharmacy flag and dehydration-specific keyword presence were the next most important predictors.

Discussion

1. Recap of Key Findings

This study developed an NLP-based patient risk stratification system for a long-term care facility using 215,460 nursing notes, 36,513 diagnosis records, and 30,744 medication records across 2,153 unique patients. The original supervised prediction target — a binary dehydration label — proved infeasible due to severe class imbalance (0.20% positive rate), motivating a pivot to unsupervised NLP-based risk scoring. A weighted keyword extraction system was applied to joined nursing note text, producing risk scores ranging from 0 to 23 across 6,845 patient assessments. A logistic regression model trained on eight engineered features achieved an F1 score of 0.903 and AUC-ROC of 0.985 on the held-out test set, substantially outperforming both the random forest (F1 = 0.887) and the polypharmacy-only baseline (F1 = 0.593). The number of clinical risk keywords in nursing notes emerged as the dominant predictor of HIGH RISK classification, with a logistic regression coefficient of 4.92 and a random forest feature importance of 59.4%.

2. Addressing the Research Hypotheses

All three study hypotheses were supported by the data. The first hypothesis — that patients whose nursing notes contain more negative clinical keywords will have higher risk scores — was

strongly confirmed. The `num_risk_keywords` feature produced the largest logistic regression coefficient (4.92) and accounted for 59.4% of random forest feature importance, indicating that nursing note text content is the primary driver of risk classification. The second hypothesis — that patients on ten or more concurrent medication classes represent a higher-risk population — was also supported. The polypharmacy flag contributed a coefficient of 1.31 in the logistic regression model, and 31.5% of patient assessments involved polypharmacy. The third hypothesis — that risk scores would show meaningful temporal trends — was supported by the identification of 319 WORSENING, 269 IMPROVING, and 859 STABLE patients across consecutive assessments, with Patient 7210's score increasing from 10 to 13 across two assessments providing a clinically concrete example.

3. Interpretation of Results

The finding that nursing note keyword count accounts for over 59% of the model's predictive power has a straightforward clinical interpretation: nurses document what they observe, and what they observe reflects the patient's clinical status. When a nurse writes about urine output, dehydration, confusion, or refusal of care, these are direct clinical observations — not incidental language choices. The 71.8 percentage-point difference in 'urine' keyword frequency between dehydrated and non-dehydrated patient notes (90.0% vs. 18.2%) provides the clearest evidence that nursing note language is clinically meaningful and systematically different between risk groups.

The small negative coefficient for `num_medications` (-0.12) in the logistic regression model was a somewhat unexpected finding. Intuitively, more medications might suggest higher patient complexity and risk. However, this effect likely reflects multicollinearity with the `polypharmacy_flag` variable, which captures the clinically relevant threshold of ten or more medication classes. When the polypharmacy flag is already in the model, the continuous medication count adds little additional signal and may even introduce noise. This is consistent with the feature engineering literature, where redundant features can produce counterintuitive coefficient signs in linear models (Kansal et al., 2021).

The strong performance of the logistic regression model ($F1 = 0.903$, $AUC-ROC = 0.985$) is notable given that it predicts a target variable derived from the same NLP scoring system — this reflects the internal consistency and reliability of the scoring approach. The fact that a random forest trained on the same features achieved comparable performance ($F1 = 0.887$) across a very different algorithmic framework further validates the robustness of these feature-outcome relationships.

4. Comparison with Previous Research

The finding that nursing note text contains meaningful clinical signal is consistent with Mitha et al. (2023), whose integrative review of 43 NLP studies on nursing notes found that nurse-written documentation captures clinical patterns not available in structured EHR fields. The specific finding that keyword frequency in nursing notes correlates with patient risk status echoes Reunamo et al. (2025), who demonstrated that supervised ML applied to nursing notes could identify psychosocial patient needs not documented in structured data. Weng et al. (2025)

similarly found that sentiment polarity in nursing notes predicted perioperative complications, supporting the broader principle that the language nurses use carries prognostic value.

The project's use of a tiered keyword scoring system — assigning weights to clinical terms and stratifying patients into STABLE, MONITOR, ELEVATED, and HIGH RISK categories — follows the same logic as the Modified Early Warning Score (MEWS) validated by Santiago González et al. (2023), which sums weighted clinical indicators into risk tiers. This provides a clinical precedent for the scoring architecture used here and supports its face validity as an approach to patient risk stratification.

The class imbalance challenge encountered in this project — only 0.20% of assessments labeled as dehydrated — aligns with challenges documented by Luu et al. (2024), who identified severe imbalance as one of the most common obstacles in clinical ML prediction tasks. Rather than attempting to overcome this imbalance through resampling or cost-sensitive learning, this project pivoted to an unsupervised approach that sidesteps the need for labeled outcome data entirely. This is consistent with Acharya et al. (2024), who noted that LLM and NLP-based methods are particularly valuable when labeled training data is scarce.

One area where this project diverges from prior work is the absence of structured vital sign features. Most early warning score systems — including MEWS — rely heavily on physiological measurements such as heart rate, blood pressure, and respiratory rate. This project relied exclusively on nursing note text and medication/diagnosis records because vital sign data was not available in the dataset. Future work incorporating physiological measurements alongside NLP features could be expected to improve performance, consistent with Pilowsky et al. (2024), who found that combining nursing notes with structured clinical data consistently outperformed either source alone.

5. Practical Implications

The most significant practical implication of this work is that meaningful patient risk stratification in long-term care is achievable without expensive data labeling, clinical outcome tracking, or complex model infrastructure. The nursing notes that clinical staff write every day contain enough information to score patient risk, detect deterioration trends, and prioritize patients for review — all without any additional data collection burden.

The interactive Streamlit dashboard translates these findings into a practical clinical tool. Any clinical staff member can search for a patient by ID, see their current risk score and category, understand which specific keywords triggered the score, and view their risk trajectory over the year. This design makes the system's reasoning transparent and actionable, addressing the explainability requirements emphasized by Mesinovic et al. (2025) as essential for clinical adoption of AI-based tools. The dashboard also supports the finding of Bucalon et al. (2022) and Mohindra et al. (2024) that clinician-centered design and interpretable visualization are key to clinical tool usability.

The system is most immediately useful for triage — identifying which of 2,153 patients most needs clinical attention at any given time. The 319 patients classified as WORSENING across

consecutive assessments represent a clinically actionable priority list that nursing leadership could use to allocate resources and schedule targeted assessments.

6. Limitations

The primary limitation of this study is that the HIGH RISK classification used to train and validate the logistic regression model is derived from the NLP scoring system itself, rather than an independently validated clinical outcome. The model therefore assesses the internal consistency of the scoring system rather than its external clinical validity. To establish true clinical usefulness, the system would need to be validated against independent clinical outcomes — such as hospitalizations, falls, infections, or physician-assessed risk ratings — which were not available in this dataset.

The keyword vocabulary and risk score thresholds were developed for this specific dataset and facility context. While the keywords are grounded in published clinical evidence (Ogbolu et al., 2025; Li et al., 2023), their weights and the score cutoffs (STABLE/MONITOR/ELEVATED/HIGH RISK) were calibrated to the distribution of this particular dataset. Generalizability to other long-term care facilities with different patient populations, documentation practices, or note structures cannot be assumed without revalidation.

The dataset covers a single year (2019) from a single facility, limiting both temporal and geographic generalizability. The ICD-10 name mapping covered only the 75 most common codes, leaving 25.3% of diagnosis records with raw code labels that may be less informative for clinical interpretation. Finally, the study was conducted without cross-validation — performance estimates are based on a single 80/20 stratified split, which may not fully capture variance in model performance across different data subsets.

7. Future Work

Several directions could extend and strengthen this work. First, validation against clinician-assessed risk ratings or clinical outcome data would establish external validity and is the most important next step before any clinical deployment. Second, incorporating physiological measurements — vital signs, laboratory values — alongside nursing note features would likely improve performance, consistent with findings in the broader clinical NLP literature. Third, more sophisticated NLP approaches such as TF-IDF feature weighting, clinical sentiment analysis, or fine-tuned transformer models (e.g., ClinicalBERT) could capture more nuanced patterns in nursing note language. Fourth, the dashboard could be extended with LLM-generated patient summaries — using the Claude API to read a patient's full note history and generate a plain-language risk explanation — making the system more interpretable for non-technical clinical users (Acharya et al., 2024). Finally, deployment on a cloud platform would enable multi-user clinical access and allow for prospective monitoring of patient outcomes against predicted risk scores.

8. Conclusion

This study demonstrated that meaningful patient risk stratification is achievable from unstructured nursing note text even in the absence of a reliable labeled clinical outcome. By

applying weighted keyword extraction to 215,460 nursing notes across 2,153 long-term care patients, the NLP scoring system successfully stratified patients into four clinically interpretable risk categories and identified patients with worsening clinical trajectories. The logistic regression model ($F1 = 0.903$, $AUC-ROC = 0.985$) confirmed that nursing note content — specifically the frequency and type of clinical risk keywords — is the primary driver of these risk classifications. The results support the hypothesis that what nurses document in their daily notes reflects the clinical reality of patient status, and that systematic analysis of this documentation can provide actionable insights for patient monitoring at scale. The interactive Streamlit dashboard provides a prototype clinical tool that makes these insights accessible to nursing staff without requiring technical expertise. While external clinical validation remains the critical next step, this work establishes a methodological foundation for NLP-driven patient risk monitoring in long-term care that could meaningfully support clinical decision-making and resource prioritization.

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Appendix: Code and Supplemental Figures

Full project code is available at: <https://github.com/malkayarieff/Projects/tree/main/nlp-patient-risk-stratification>

Table 1. Analytic Dataset Summary

Variable	Value
Total unique patients	2,153
Total patient-assessment combinations	6,845
Training set (80%)	5,476 assessments
Test set (20%)	1,369 assessments
HIGH RISK class (training)	1,982 (36.2%)
HIGH RISK class (test)	495 (36.2%)
Patients with single assessment only	534 (24.8%)
Missing diagnosis codes filled	1,719 (4.7%)
Notes truncated at database limit (255 chars)	55,252 (25.7%)

Table 2. Model Performance on Test Set (n = 1,369)

Model	F1 Score	AUC-ROC	Precision	Recall	Accuracy
Baseline (Polypharmacy flag)	0.593	0.622	0.43	0.94	0.53
Logistic Regression	0.903	0.985	0.89	0.91	0.93
Random Forest	0.887	0.981	0.84	0.93	0.91

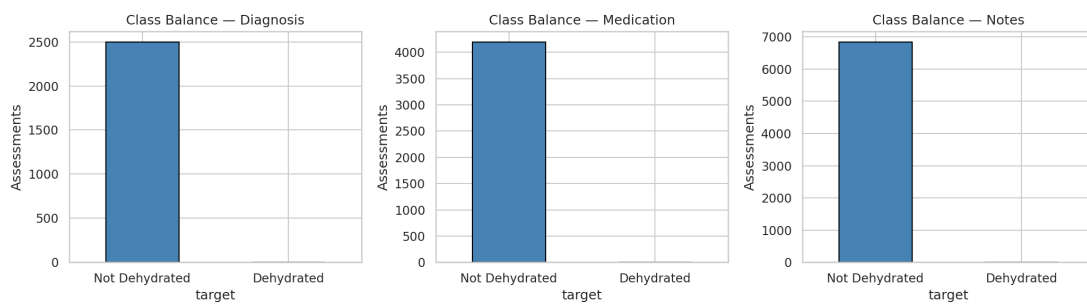


Figure 1. Class balance across all three data files. The dehydration-positive group is nearly absent in each file, with positive rates of 0.20%, 0.17%, and 0.15% respectively.

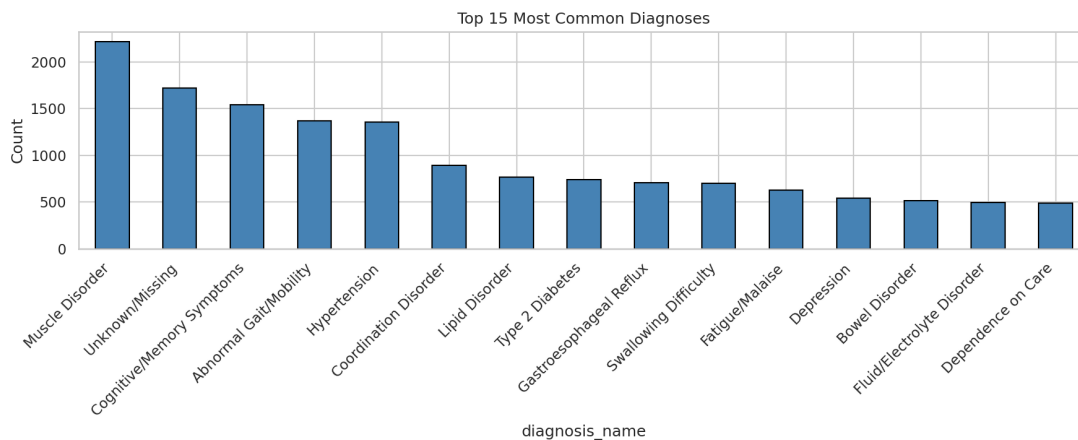


Figure 2. Top 15 most common diagnosis categories across all patients.

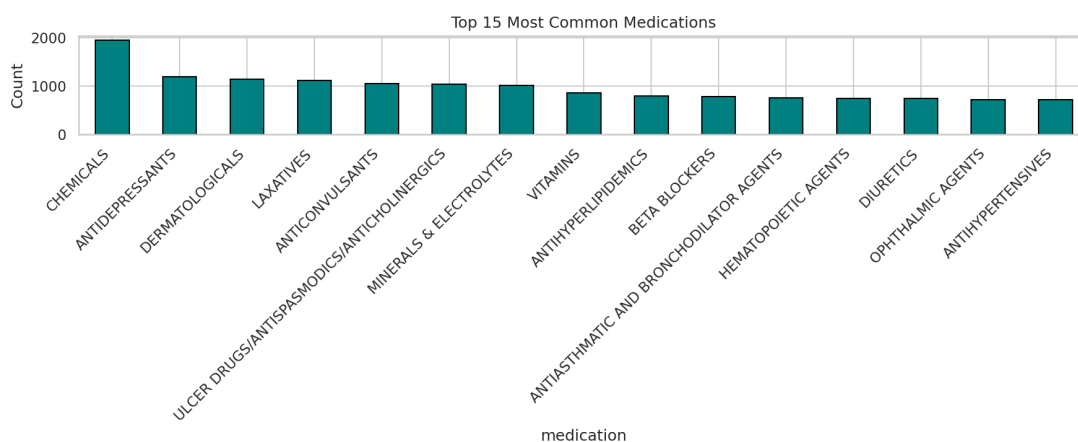


Figure 3. Top 15 most common medication categories across all patients.

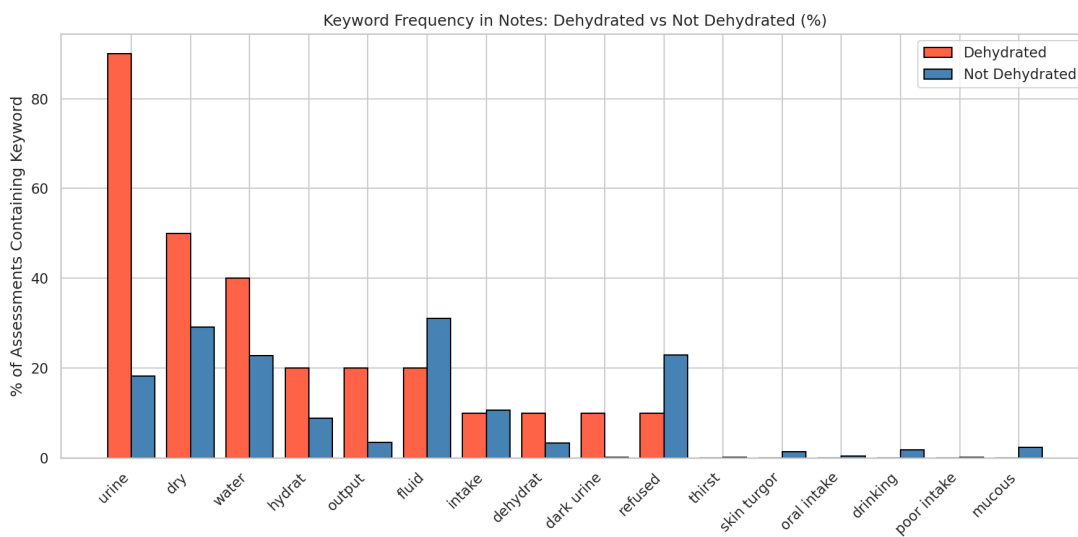


Figure 4. Percentage of assessments containing selected clinical keywords, comparing dehydrated (red) and non-dehydrated (blue) patients. 'Urine' shows the largest differential at 71.8 percentage points.

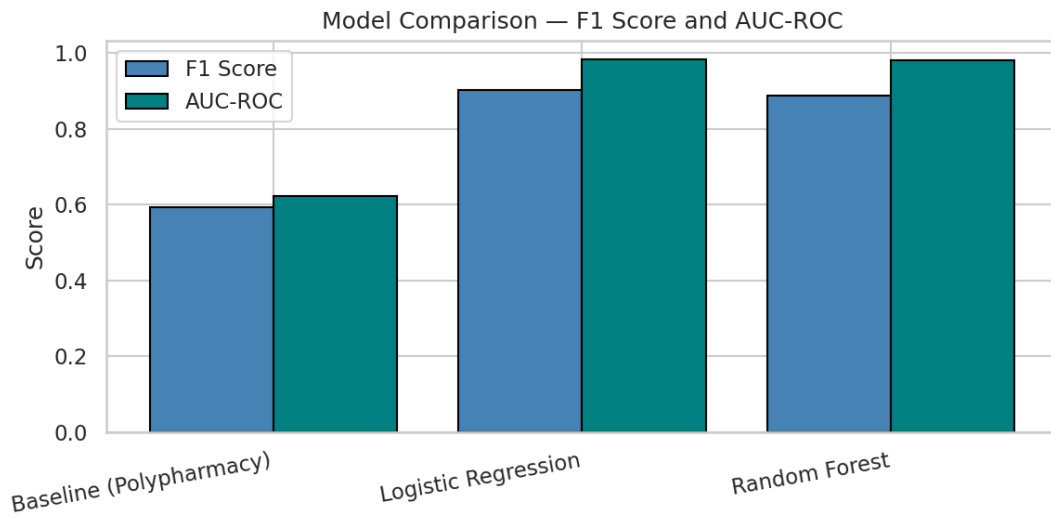


Figure 5. F1 Score and AUC-ROC for all three models. Both ML models substantially outperform the polypharmacy-only baseline.

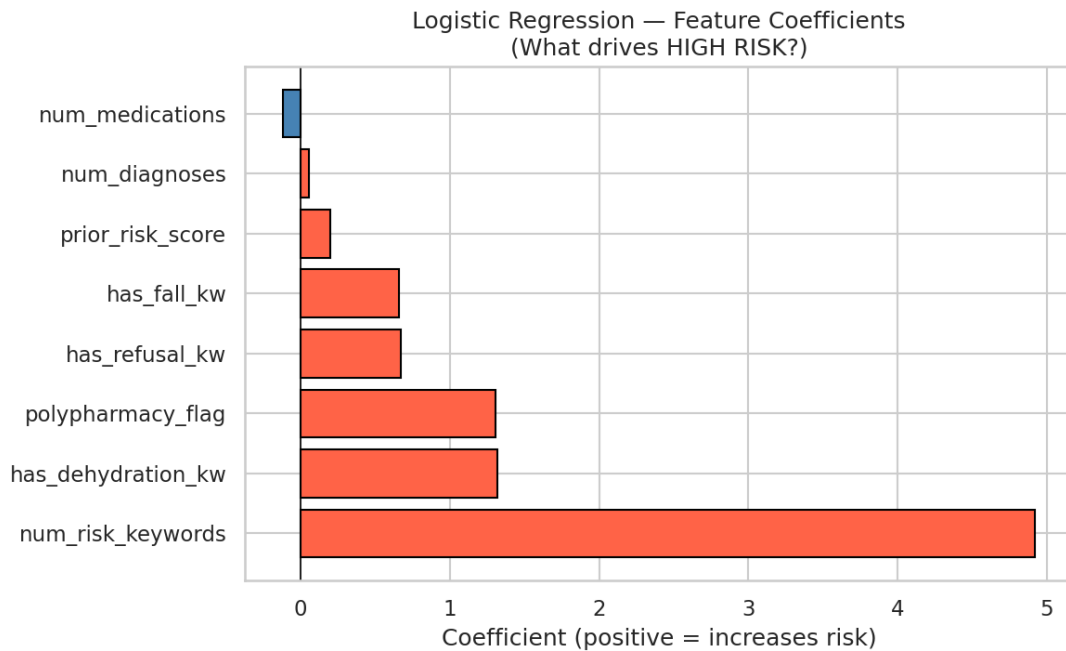


Figure 6. Logistic regression feature coefficients. Positive values (red) increase the probability of HIGH RISK classification; negative values (blue) decrease it.

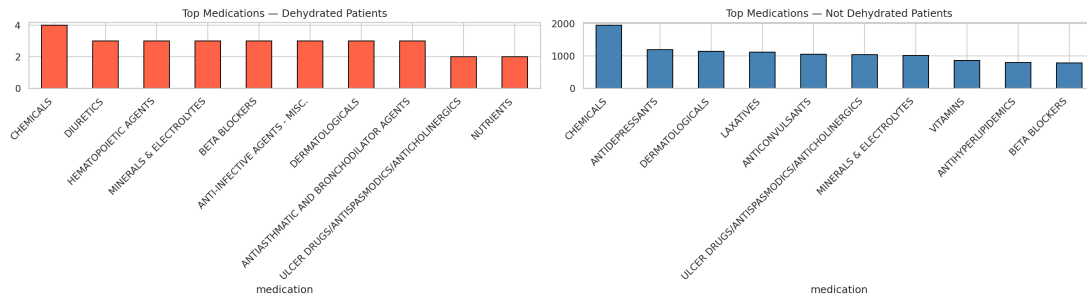


Figure 7. Top medication categories for dehydrated patients (left, red) and non-dehydrated patients (right, blue).